

Normal Distribution

Theory, Solved Problems and Normal Curves

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Probability and Random Processes • Unit I

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Theory

Binomial $\longrightarrow$ Normal Distribution

In $B(n, p)$, when **neither p nor q is small**, i.e. np is large:

Stirling's formula: $n! = n^n e^{-n} \sqrt{2\pi n}$

$$f(x) = \binom{n}{x} p^x q^{n-x} \approx \frac{1}{\sqrt{2\pi npq}} e^{-\frac{(x-np)^2}{2npq}}$$

Since $\mu = np$ and $\sigma^2 = npq$,

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$P(x_1 \leq X \leq x_2) = \int_{x_1}^{x_2} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

Normal or Gaussian Distribution

Definition

A CRV X follows the **normal (Gaussian)** distribution if its PDF is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty$$

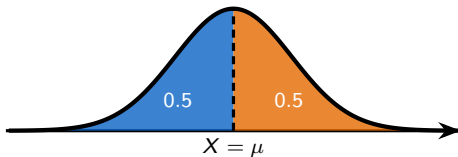
and we write $X \sim N(\mu, \sigma^2)$.

Results

(i) MGF = $e^{\mu t + \frac{\sigma^2 t^2}{2}}$ (ii) Mean = μ and Variance = σ^2

Properties of the Normal Curve

1. Symmetric about $X = \mu$ and bell shaped.
2. Total area under the curve is **unity**.
3. Area on either side of $X = \mu$ is **0.5**.



Standard Normal Variate

Let $X \sim N(\mu, \sigma^2)$ and put

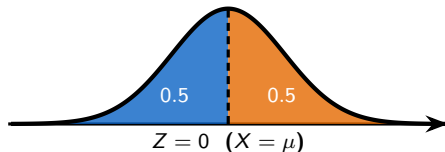
$$\boxed{Z = \frac{X - \mu}{\sigma}} \implies E(Z) = 0, \quad \text{Var}(Z) = 1$$

so that $Z \sim N(0, 1)$ is the **standard normal variate**, with PDF

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty.$$

The curve is symmetric about $Z = 0$ and bell shaped.

Areas under the Standard Normal Curve



$$P(-\infty < Z < 0) = 0.5$$

$$P(0 < Z < \infty) = 0.5$$

$$P(-\infty < Z < \infty) = 1$$

By symmetry,

$$P(-a < Z < 0) = P(0 < Z < a)$$

Normal tables give $P(0 < Z < a)$.

Problem 1 — Heights of Soldiers

Problem 1

Question

Assume that the mean height of soldiers is 68.22 inches with variance 10.88 inches. How many soldiers in a regiment of 2000 soldiers would you expect to be over 6 feet tall? Assume heights are normally distributed.

Problem 1 — Solution (Step 1: Data)

Let $X \rightarrow$ height of soldiers, $X \sim N(\mu, \sigma^2)$

$$\mu = 68.22 \text{ inches}, \quad \sigma^2 = 10.88, \quad \sigma = \sqrt{10.88} = 3.2985$$

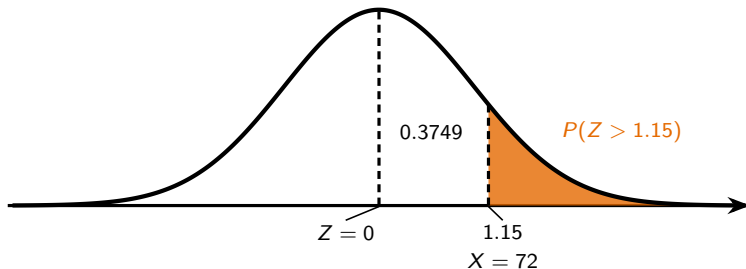
Since 6 feet = 72 inches,

$$P(\text{height more than 6 feet}) = P(X > 72)$$

Problem 1 — Solution (Step 2: Standardise)

$$\begin{aligned} P(X > 72) &= P\left(\frac{X - \mu}{\sigma} > \frac{72 - \mu}{\sigma}\right) \\ &= P\left(Z > \frac{72 - 68.22}{\sqrt{10.88}}\right) = P\left(Z > \frac{3.78}{3.2985}\right) \\ &= P(Z > 1.146) \approx P(Z > 1.15) \end{aligned}$$

Problem 1 — Solution (Step 3: Normal Curve)



$$P(Z > 1.15) = 0.5 - P(0 < Z < 1.15) = 0.5 - 0.3749 = 0.1251$$

Problem 1 — Solution (Step 4: Answer)

$$P(X > 6 \text{ feet}) = 0.1251$$

Number of soldiers, in a regiment of 2000, having height more than 6 feet

$$= 2000 \times 0.1251 = 250.2$$

Answer

$\approx$ 250 soldiers

Problem 2 — Mean and SD from Percentages

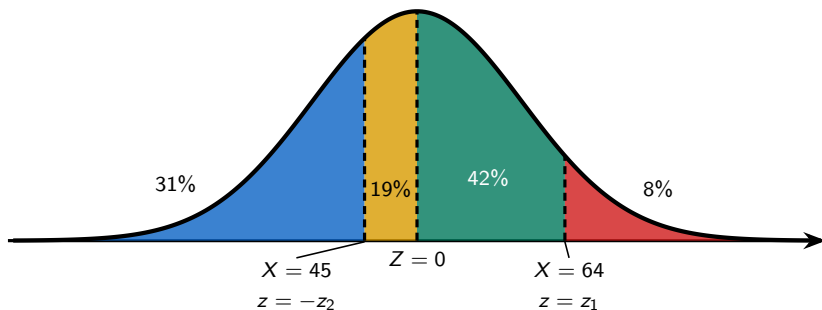
Problem 2

Question

In a normal distribution 31% of the items are under 45 and 8% of the items are over 64. Find the mean and standard deviation of the distribution.

Problem 2 — Solution (Step 1: The Picture)

Given $X \sim N(\mu, \sigma^2)$ with $P(X < 45) = 0.31$ and $P(X > 64) = 0.08$. Since $31\% < 50\%$, the value 45 lies to the **left** of μ ; since $8\% < 50\%$, the value 64 lies to the **right** of μ .



Problem 2 — Solution (Step 2: Read the z values)

Let $Z = z_1$ and $Z = -z_2$ be the points corresponding to $X = 64$ and $X = 45$. From the figure:

$$P(0 < Z < z_1) = 42\% = 0.42 \implies \boxed{z_1 = 1.41}$$

$$P(-z_2 < Z < 0) = 19\% \implies P(0 < Z < z_2) = 0.19 \text{ (symmetry)}$$

$$\implies \boxed{z_2 = 0.50}$$

(both from the normal table)

Problem 2 — Solution (Step 3: Two Equations)

Using $Z = \frac{X - \mu}{\sigma}$:

$$z_1 = \frac{64 - \mu}{\sigma}$$

$$1.41 = \frac{64 - \mu}{\sigma}$$

$$\boxed{\mu + 1.41 \sigma = 64} \quad (1)$$

$$-z_2 = \frac{45 - \mu}{\sigma}$$

$$-0.50 = \frac{45 - \mu}{\sigma}$$

$$\boxed{\mu - 0.50 \sigma = 45} \quad (2)$$

Problem 2 — Solution (Step 4: Solve)

(1) – (2):

$$1.91 \sigma = 19 \implies \sigma = 9.9476$$

Substituting in (2):

$$\mu = 45 + 0.50(9.9476) = 45 + 4.974 = 49.97$$

Answer

Mean = $\mu \approx 50$ and S.D. = $\sigma \approx 10$

Problem 3 — Heights of a Group of Men

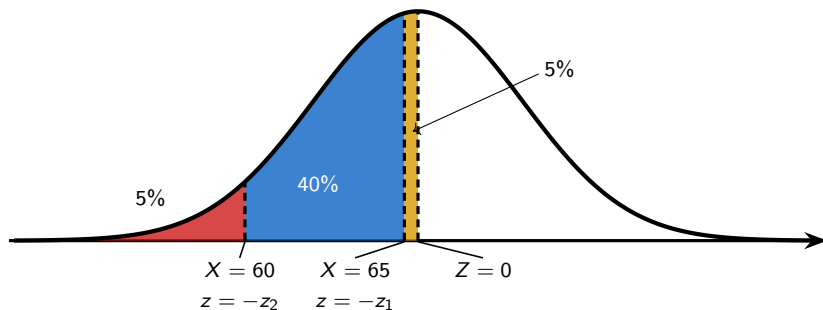
Problem 3

Question

Of a large group of men, 5% are under 60 inches in height and 40% are between 60 and 65 inches. Assuming a normal distribution, find the mean height and the standard deviation.

Problem 3 — Solution (Step 1: The Picture)

Given $P(X < 60) = 0.05$ and $P(60 < X < 65) = 0.40$, so $P(X < 65) = 0.45 < 0.5$. Hence **both** 60 and 65 lie to the left of μ .



Problem 3 — Solution (Step 2: Read the z values)

Let $Z = -z_1$ and $Z = -z_2$ be the values corresponding to $X = 65$ and $X = 60$. From the figure:

$$\begin{aligned}P(-z_1 < Z < 0) &= 5\% \\ \Rightarrow P(0 < Z < z_1) &= 0.05 \\ \Rightarrow \boxed{z_1 = 0.13}\end{aligned}$$

$$\begin{aligned}P(-z_2 < Z < 0) &= 45\% \\ \Rightarrow P(0 < Z < z_2) &= 0.45 \\ \Rightarrow \boxed{z_2 = 1.65}\end{aligned}$$

Problem 3 — Solution (Step 3: Two Equations)

By the relation $Z = \frac{X - \mu}{\sigma}$:

$$-z_1 = \frac{65 - \mu}{\sigma}$$

$$-0.13 = \frac{65 - \mu}{\sigma}$$

$$\boxed{\mu - 0.13 \sigma = 65} \quad (1)$$

$$-z_2 = \frac{60 - \mu}{\sigma}$$

$$-1.65 = \frac{60 - \mu}{\sigma}$$

$$\boxed{\mu - 1.65 \sigma = 60} \quad (2)$$

Problem 3 — Solution (Step 4: Solve)

(1) – (2):

$$1.52 \sigma = 5 \implies \sigma = 3.289$$

Substituting in (1):

$$\mu = 65 + 0.13(3.289) = 65 + 0.4276 = 65.43$$

Answer

Mean height = 65.42" S.D. = 3.29"

Problem 4 — Examination Grades

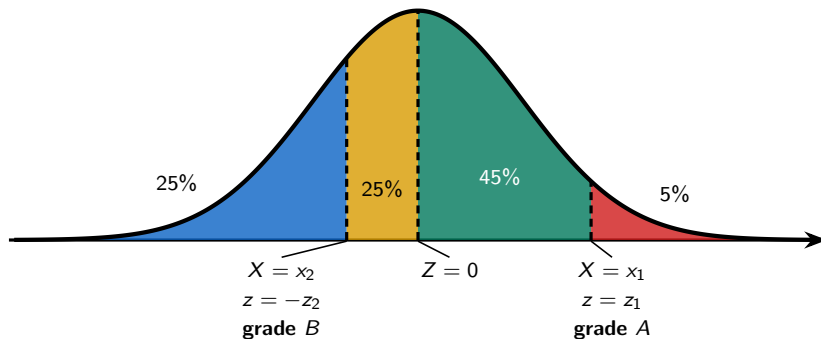
Problem 4

Question

A set of examination marks is approximately normally distributed with mean 75 and S.D. 5. If the top 5% of students get grade A and the bottom 25% get grade B , what mark is the lowest of A and what mark is the highest of B ?

Problem 4 — Solution (Step 1: The Picture)

$X \rightarrow$ examination marks, $X \sim N(\mu, \sigma^2)$ with $\mu = 75$, $\sigma = 5$.



Problem 4 — Solution (Step 2: Read the z values)

Let $X = x_1$ and $X = x_2$ be the lowest mark of grade A and the highest mark of grade B ; let $Z = z_1$ and $Z = -z_2$ be the corresponding points.

$$P(0 < Z < z_1) = 45\% = 0.45 \quad \implies \quad \boxed{z_1 = 1.65}$$

$$P(-z_2 < Z < 0) = 25\% \implies P(0 < Z < z_2) = 0.25 \implies \boxed{z_2 = 0.68}$$

(both from the normal table)

Problem 4 — Solution (Step 3: Back-transform)

Using $Z = \frac{X - \mu}{\sigma}$:

$$z_1 = \frac{x_1 - \mu}{\sigma}$$

$$1.65 = \frac{x_1 - 75}{5}$$

$$x_1 = 75 + 8.25 = 83.25$$

$$-z_2 = \frac{x_2 - \mu}{\sigma}$$

$$-0.68 = \frac{x_2 - 75}{5}$$

$$x_2 = 75 - 3.40 = 71.60$$

Answer

Lowest mark of grade A ≈ 83

Highest mark of grade B ≈ 72

Problem 5 — Examination Divisions

Problem 5

Question

In an examination a student passes if he secures 30% or more marks. He is placed in the first, second and third divisions if he secures between 60% and 80%, between 45% and 60%, and between 30% and 45% respectively. He gets a distinction if he secures 80% or more. It is noticed that 10% of the students failed and 5% obtained a distinction. Calculate the percentage of students placed in the **second division**.

Problem 5 — Solution (Step 1: Data)

Let $X \rightarrow$ marks obtained, $X \sim N(\mu, \sigma^2)$.

Failed $\Rightarrow$ marks below 30:

$$P(X < 30) = 10\% = 0.10$$

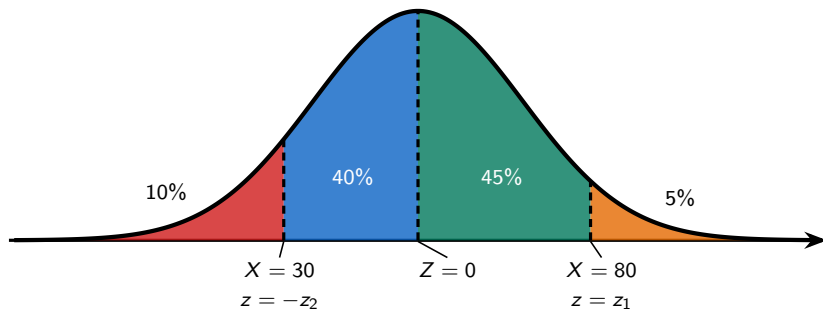
Distinction $\Rightarrow$ marks above 80:

$$P(X > 80) = 5\% = 0.05$$

Second division $\Rightarrow 45 < X < 60$ (this is what we must find).

Problem 5 — Solution (Step 2: The Picture)

Since $10\% < 50\%$, $X = 30$ lies to the left of μ ; since $5\% < 50\%$, $X = 80$ lies to the right of μ .



Problem 5 — Solution (Step 3: Read the z values)

Let $Z = -z_2$ and $Z = z_1$ correspond to $X = 30$ and $X = 80$.

$$P(-z_2 < Z < 0) = 0.40$$

$$P(0 < Z < z_1) = 0.45$$

$$\Rightarrow P(0 < Z < z_2) = 0.40$$

$$\Rightarrow \boxed{z_1 = 1.645}$$

$$\Rightarrow \boxed{z_2 = 1.28}$$

(from the normal table)

Problem 5 — Solution (Step 4: Find μ and σ)

Using $Z = \frac{X - \mu}{\sigma}$:

$$-1.28 = \frac{30 - \mu}{\sigma}$$

$$\boxed{\mu - 1.28\sigma = 30} \quad (1)$$

$$1.645 = \frac{80 - \mu}{\sigma}$$

$$\boxed{\mu + 1.645\sigma = 80} \quad (2)$$

(2) - (1):

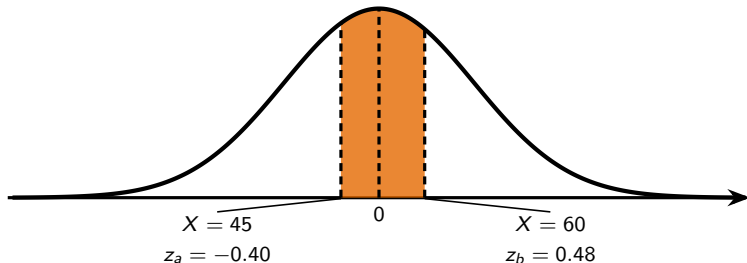
$$2.925\sigma = 50 \implies \sigma = 17.094$$

$$\mu = 30 + 1.28(17.094) = 30 + 21.88 = 51.88$$

Problem 5 — Solution (Step 5: Standardising)

Second division $\Rightarrow P(45 < X < 60)$, with $\mu = 51.88$, $\sigma = 17.094$:

$$z_a = \frac{45 - 51.88}{17.094} = \frac{-6.88}{17.094} = -0.40, \quad z_b = \frac{60 - 51.88}{17.094} = \frac{8.12}{17.094} = 0.48$$



Problem 5 — Solution (Step 6: Answer)

$$\begin{aligned}P(45 < X < 60) &= P(-0.40 < Z < 0.48) \\&= P(-0.40 < Z < 0) + P(0 < Z < 0.48) \\&= P(0 < Z < 0.40) + P(0 < Z < 0.48) \quad (\text{by symmetry}) \\&= 0.1554 + 0.1844 = 0.3398\end{aligned}$$

Answer

Students placed in the second division $= 0.3398 \times 100 \approx 34\%$

Problem 6 — Life of Dry Battery Cells

Problem 6

Question

A sample of 100 dry battery cells tested to find the length of life produced the following results:

$$\bar{x} = 12 \text{ hours}, \quad \sigma = 3 \text{ hours}$$

Assuming the data to be normally distributed, what percentage of battery cells are expected to have life

(i) more than 15 hours (ii) less than 6 hours (iii) between 10 and 14 hours?

(U.P. III Semester, Dec. 2003)

Problem 6 — Solution (Step 1: Data)

Let $X \rightarrow$ length of life of a dry battery cell, $X \sim N(\mu, \sigma^2)$

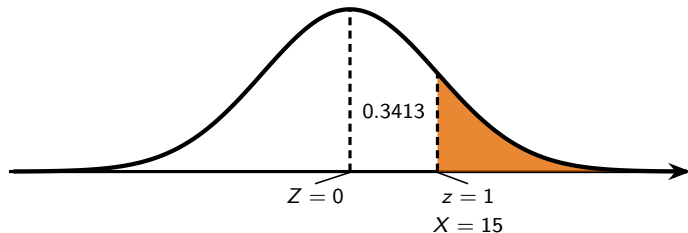
$$\mu = \bar{x} = 12 \text{ hours}, \quad \sigma = 3 \text{ hours}$$

The standard normal variate is

$$Z = \frac{X - \mu}{\sigma} = \frac{X - 12}{3}$$

Problem 6 (i) — Life more than 15 hours

When $X = 15$: $z = \frac{15 - 12}{3} = 1$

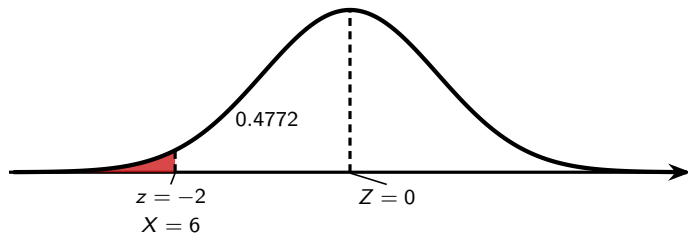


$$P(X > 15) = P(Z > 1) = 0.5 - P(0 < Z < 1) = 0.5 - 0.3413 = 0.1587$$

$\Rightarrow$ 15.87% of the cells last more than 15 hours.

Problem 6 (ii) — Life less than 6 hours

$$\text{When } X = 6 : \quad z = \frac{6 - 12}{3} = -2$$

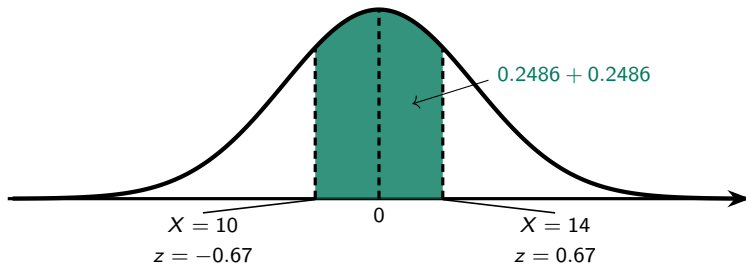


$$\begin{aligned} P(X < 6) &= P(Z < -2) = 0.5 - P(-2 < Z < 0) = 0.5 - P(0 < Z < 2) \\ &= 0.5 - 0.4772 = 0.0228 \quad \Rightarrow \quad 2.28\% \end{aligned}$$

Problem 6 (iii) — Life between 10 and 14 hours

$$X = 10 : z = \frac{10 - 12}{3} = -0.67$$

$$X = 14 : z = \frac{14 - 12}{3} = 0.67$$



Problem 6 (iii) — Answer

$$\begin{aligned}P(10 < X < 14) &= P(-0.67 < Z < 0.67) \\&= P(-0.67 < Z < 0) + P(0 < Z < 0.67) \\&= 2 P(0 < Z < 0.67) \quad (\text{by symmetry}) \\&= 2 \times 0.2486 = 0.4972\end{aligned}$$

Answer

49.72% of the battery cells have a life between 10 and 14 hours.

Summary

Summary of Answers

No.	Problem	Answer
1	Soldiers over 6 feet (out of 2000)	≈ 250
2	Mean and S.D. (31% under 45, 8% over 64)	$\mu \approx 50, \sigma \approx 10$
3	Mean height and S.D. of men	$\mu = 65.42'', \sigma = 3.29''$
4	Lowest mark of A / highest mark of B	83 / 72
5	Students in the second division	$\approx 34\%$
6	Battery life: $> 15\text{h}$ / $< 6\text{h}$ / 10–14h	15.87% / 2.28% / 49.72%

Thank you